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1 - ABSTRACT

Lipid polymorphism is the most widely reported theory for the instability of drug release after the storage of solid lipid dosage forms. The present work gives new insights into the real-time polymorphic changes of Compritol 888 in tablets prepared by direct compression. A non-destructive analytical method was used allowing correlation of the type of lipid polymorph with the respective drug release kinetics prior to and following storage at 40°C. The findings indicate a new theory behind the sometimes encountered changes in drug release profile during storage.

2 - INTRODUCTION

Solid lipid excipients are crystalline materials which exhibit different polymorphs¹. Used as sustained release (SR) matrix formers, the lipid crystal lattice and inherent polymorphs can change over time, potentially affecting the release kinetics of oral solid dosage forms *in vitro*. Lipid polymorphism is therefore the most widely reported reason for storage instability of SR solid lipid dosage forms². To our knowledge however, no direct correlation between polymorphic evolution of the lipid excipient and drug release has been reported yet. Given numerous functional advantages of solid lipid excipients it is worthwhile investigating the real impact of lipid polymorphism on drug release stability using new techniques such as micro x-ray diffraction. This method enabled analysis of the polymorphic state of Compritol 888 within intact tablets before and after storage, and the correlation of the lipid polymorph with the respective drug release kinetics for the first time.

3 - EXPERIMENTAL METHODS

20% theophylline, 15% glyceryl behenate (Compritol® 888 ATO), 32.25% DCPA (Fujicalin SG), 32.25% lactose (Tablettose 80) and 0.5% magnesium stearate was compressed into 12mm flat tablets using a RIVA Piccola rotary press at 30rpm and 18kN. Tablets (uncured and cured for 7 days at 50°C) were stored in closed glass vials at 40°C/75% RH. The Compritol 888 polymorph was determined by micro X-ray diffraction. The exact set-up is described by Jannin et al.³

4 - RESULTS AND DISCUSSION

Micro X-ray diffraction allowed elucidation of the actual polymorphic state of Compritol 888 in a tablet formulation and correlation of the lipid polymorph with the drug release kinetics. The bars on the x-axes in Figure 1 represent the diffraction peaks of Compritol 888 (polymorph) before and after 45 days of storage for uncured and cured tablets. The respective release profiles are shown below.

Drug release from uncured tablets decreased after 45 days of storage although no changes in the polymorph of Compritol 888 were observed (blue bars). In contrast, the polymorph of Compritol 888 in cured tablets changed (although insignificantly, red bars) whilst drug release after storage was stable.

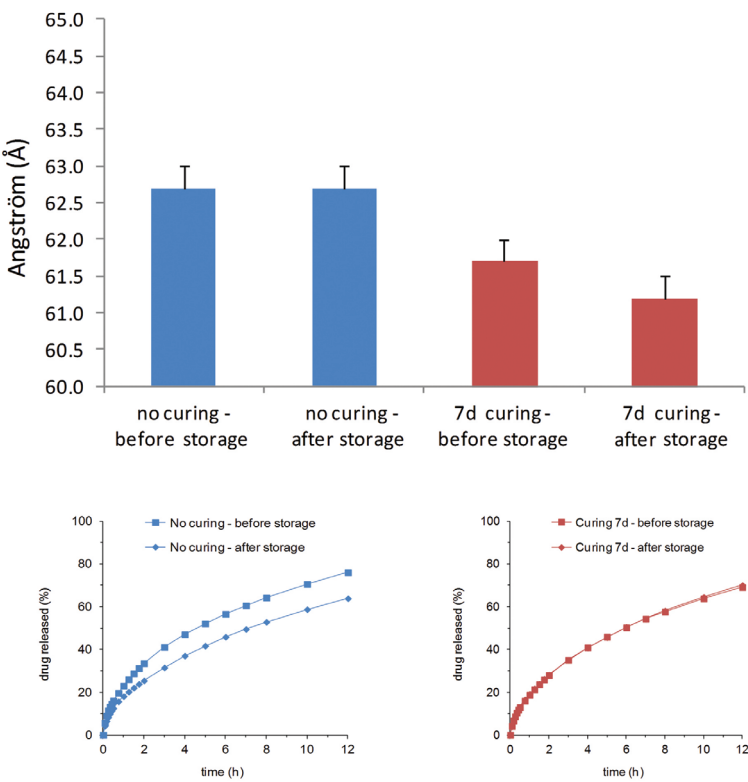


Figure 1: Correlation of the real-time polymorphic state of Compritol 888 with the drug release kinetics before and after 45 days storage.

Within the scope of this study we demonstrate that an evolution of the lipid polymorph is not necessarily linked to drug release changes over time.

Therefore several hypotheses can be envisaged to explain the observed changes of drug release kinetics after storage, including: (i) evolution of the physicochemical properties of the tablet, (ii) changes in matrix porosity or (iii) wettability of the tablets.

Table 1 summarizes the physicochemical properties of the tablets before and after storage for 45 days. The crushing strength slightly decreased for uncured tablets, and increased for the cured tablets. Nevertheless, the differences in tablet hardness could not be correlated to drug release variations (data not shown). Tablet weight and thickness remained unchanged during storage and do therefore not affect drug release (Table 1).

Table 1: Physicochemical tablet properties.

Tablet	Weight (mg)	Thickness (mm)	Tensile strength (MPa)
No curing - before storage	503 ± 4.1	3.23 ± 0.02	1.55 ± 0.08
No curing - after storage	503 ± 3.6	3.24 ± 0.03	1.45 ± 0.09
7d curing - before storage	504 ± 4.0	3.25 ± 0.02	1.21 ± 0.06
7d curing - after storage	504 ± 4.5	3.22 ± 0.03	1.38 ± 0.04

Figure 2 shows the evolution of the contact angle of a water droplet onto the tablet surface during storage for 3 months. It is clear that the wettability of the matrix is reduced after storage which can lead to the deceleration of water diffusion into the tablet and drug diffusion out of the tablet, therefore leading to slower drug release kinetics. The change of wettability can have different causes. For instance, humidity during storage (75% RH) can induce moisture saturation leading to reduced matrix wettability (Table 2).

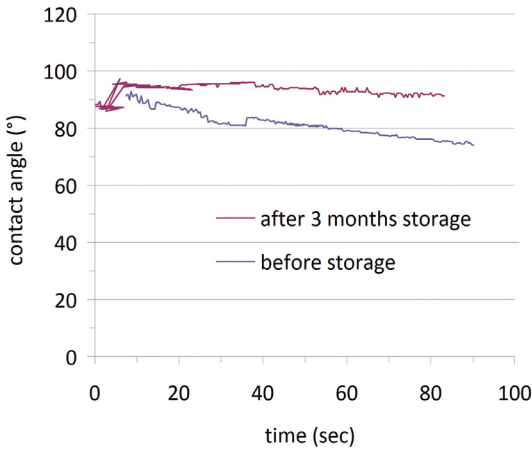


Figure 2: Wettability of the uncured tablets before and after 3 months storage.

Table 2 illustrates how the presence of humidity during storage can impact drug release from the investigated tablets. The similarity factor f2 decreased from 86 to 51 at 75% RH for the uncured tablets (Fujicalin), indicating a difference of almost 10% between the release kinetics before and after 6 months storage. Interestingly this effect was reduced or even absent when Fujicalin was replaced by the chemically identical excipient Emcompress (Table 2). The physical nature of the filler might therefore be a determining factor in release rate changes over time rather than the lipid polymorph.

Table 2: Effect of humidity on drug release after 6 months storage.

Excipient	Chemical	f2 at 40°C	f2 at 40°C/75% RH
Fujicalin SG	DCPA*	86	51
Emcompress anhydrous	DCPA	66	66
Emcompress Premium	DCP*	87	80

* DCP(A) = Dibasic calcium phosphate (anhydrous)

5 - CONCLUSION

Drug release changes after storage with solid lipid SR matrix tablets were not related to evolution of the lipid polymorph. The present study casts doubt on the widely reported theory that lipid polymorphism affects drug release over time. New hypothesis are warranted including the sensitivity of certain tablet ingredients to moisture. Further investigations will follow.

6 - REFERENCES

[1] Sato, K. Chem. Eng. Sci. 56 (2001) 2255-2265.
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[3] Jannin, V. et al. J. Control. Rel. 197 (2015) 158-164.

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